



Volume of cylinders

- 1 Volume is the amount of space occupied by a closed 3D shape. Fill in the blank
- 2 Which of the following is the formula for the volume of a cylinder, where r is the radius and h is the height? Tick **1** correct answer
- ☐ $\pi r^3 h$
- ☒ $\pi r^2 h$
- ☐ $\pi r h^2$
- ☐ $r^2 h$
- 3 A cylinder with a radius of 6 cm and a height of 10 cm, has a volume of 360 $\pi \text{ cm}^3$. Fill in the blank
- 4 If the radius of a cylinder was to stay the same but the height was to change, what would happen to the volume? Tick **1** correct answer
- ☐ increase
- ☐ decrease
- ☐ stay the same
- ☒ you cannot tell



- 5 A cylinder has a radius of 9 cm and a volume of $810\pi \text{ cm}^3$. What is the length of the cylinder? Tick **1** correct answer
- ☐ $10\pi \text{ cm}$
- ☒ 10 cm
- ☐ $90\pi \text{ cm}$
- ☐ 90 cm
- 6 If you know the volume of a cylinder and its length, what other information can you work out? Tick **3** correct answers
- ☒ The radius of the circular face.
- ☒ The area of the circular face.
- ☐ The mass of the cylinder.
- ☒ The surface area of the cylinder.
- ☐ The colour of the cylinder.

